ARAGOCOR MINERALS

Fertilizer Enhancer / Granular — Technical Data

High-purity natural oolitic aragonite · Technical documentation

Technical Data

Fertilizer Enhancer

Oolitic aragonite - the world's only recent origin, renewable and sustainable performance mineral

Fertilizer Enhancer /Granular (Oolitic Aragonite)

This is a naturally precipitated, high surface area calcium carbonate in the crystalline form of aragonite. This lower carbon footprint natural material can be used in place of all limestone derived products. The higher solubility of aragonite versus calcite in combination with an extremely large surface area associated with oolites and an extremely pure material (>39% calcium) provides an extremely efficient calcium carbonate. Oolitic aragonite is a highly pure, renewable and biogenic mineral, which is ideally suited to enhance fertilizer performance. Containing approximately 40% calcium and able to dissolve quickly, aragonite is a practical and efficient pH buffer. Further, aragonite has an innate ability to absorb nutrients deliverable into fertilizers, efficiently increasing their availability. One of the delivery mechanisms is made possible because of a negative zeta potential in combination with a large surface area of the aragonite crystal. Aragonite also acts to neutralize volatilization of ammonia by reducing the hydrolysis of urea by delivering free calcium ions within fertilizers, thereby stabilizing the ammonia.

Aragonite Composition Large surface area and porosity

Properties

Oolitic Aragonite

Calcite/Limestone

Specific Gravity

2.93+

2.7

Mohs Hardness

3.5 - 4

3

Crystal Structure

orthorhombic

hexagonal

Surface Area

> 1.82m^2/g

< .55m^2/g

Zeta Potential

-33.85mV to - 6.65mV

-1.01mV to +11.55Mv

Micro porosity

Very high

Low

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